

Procurement Fraud & Supplier Risk Analysis Dashboard

SQL Appendix

Prepared By

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Tools Used

- PostgreSQL 17
 - SQL
 - Python
 - Power BI
 - pgAdmin 4
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Purpose of this Appendix

This document contains the SQL queries used to generate the key metrics, KPIs, and visualizations presented in the Procurement Fraud & Supplier Risk Dashboard.

The appendix provides technical evidence of SQL development, data analysis, and reporting skills used throughout the project.

1. KPI Analysis Queries

1.1 Total Spend KPI

Purpose

Calculate total procurement spending.

SQL Query

```
SELECT  
  
    SUM(invoice_amount) AS total_spend  
FROM analytics.invoice_risk_flags;
```

Result

£132.98K

1.2 Total Invoices KPI

Purpose

Calculate total invoices processed.

SQL Query

```
SELECT  
  
    COUNT(*) AS total_invoices
```

```
FROM analytics.invoice_risk_flags;
```

Result

```
10
```

1.3 High-Risk Suppliers KPI

Purpose

Identify suppliers exceeding the defined risk threshold.

SQL Query

```
SELECT
    COUNT(*)
FROM analytics.supplier_risk_summary
WHERE total_risk_score >= 2;
```

Result

```
3
```

1.4 High-Value Invoices KPI

Purpose

Identify invoices exceeding the high-value threshold.

SQL Query

```
SELECT
    SUM(high_value_invoice_flag)
FROM analytics.invoice_risk_flags;
```

Result

```
3
```

2. Dashboard Analysis Queries

2.1 Department Spend Analysis

SQL Query

```
SELECT
    department,
    SUM(invoice_amount) AS total_spend
FROM analytics.invoice_risk_flags
GROUP BY department
```

```
ORDER BY total_spend DESC;
```

2.2 Supplier Risk Analysis

SQL Query

```
SELECT
    supplier_name,
    total_risk_score
FROM analytics.supplier_risk_summary
ORDER BY total_risk_score DESC;
```

2.3 Top Risky Invoices

SQL Query

```
SELECT
    invoice_id,
    invoice_risk_score
FROM analytics.invoice_risk_flags
ORDER BY invoice_risk_score DESC
LIMIT 5;
```

2.4 Invoice Risk Detail Analysis

SQL Query

```
SELECT
    invoice_id,
    high_risk_supplier_flag,
    high_value_invoice_flag,
    late_payment_flag,
    missing_po_flag,
    approval_risk_flag,
    invoice_risk_score
FROM analytics.invoice_risk_flags
ORDER BY invoice_risk_score DESC;
```